

Date	07 July 2026	Team Name	Anupriya
Project Name	CaptionAI	Maximum Marks	3 Marks

PROJECT EXECUTABLE FILES & EXECUTION GUIDE

SmartBridge Skill Wallet Submission

This document details the configuration files, directory structures, and environment setup instructions required by evaluators to access, review, and locally launch the CaptionAI generation engine.

Step 1: Project Links & Access

RESOURCE TYPE	URL / ACCESS LINK
Source Code Repository	https://github.com/anu-priya2/CaptionAI-AI-Powered-Social-Media-Caption-Generator
Demo Video Link	https://drive.google.com/file/d/1MAojA5waTHLnrL9jabenEM6jC2t5ra_U/view?usp=drivesdk

Step 2: Key Executable Files & Folder Structure

S.NO	FOLDER / FILE NAME	DESCRIPTION & PURPOSE
1	/static/css	Contains the custom CSS stylesheet (<code>style.css</code>) that creates the modern dark glassmorphic user interface (UI).
2	/static/js	Contains the client-side JavaScript file (<code>main.js</code>) handling asynchronous form submissions and dynamic UI updates via the Fetch API.
3	/templates	Contains the main single-page application dashboard interface layout file (<code>index.html</code>).
4	.env	Stores sensitive environment setup keys including your private <code>GROQ_API_KEY</code> configuration variables safely.
5	app.py	The main Flask backend script managing routing configuration, input verification, prompt packaging, and Groq API calls.
6	requirements.txt	Text file listing all necessary Python external application packaging framework libraries (<code>Flask</code> , <code>groq</code> , <code>python-dotenv</code> , <code>pyngrok</code>).
7	run_public.py	Custom deployment helper execution script designed to automatically launch the background web server and active Ngrok tunnels simultaneously.

Step 3: Local Setup & Execution Instructions

STEP	ACTION / TERMINAL COMMAND	EXPECTED OUTCOME / NOTES
1	Clone or download the project source code directory into your local workspace.	Opens up the full application project folder structure directory to begin local environment staging.
2	<code>python -m venv myenv</code>	Initializes a clean, isolated virtual environment workspace container to manage project packages.
3	<code>myenv\Scripts\activate</code>	Activates the virtual environment system shell safely so package installations don't conflict with global paths (Windows Command Prompt layout).
4	<code>pip install -r requirements.txt</code>	Downloads and installs the exact versions of Flask, Groq SDK, python-dotenv, and pyngrok packages required.
5	Configure <code>.env</code> file with your key parameters and add the Ngrok token within <code>run_public.py</code> .	Sets up secure backend communication interfaces so the application can communicate smoothly with the Groq AI models.
6	<code>python run_public.py</code>	Automatically spins up the local backend server engine and generates your secure public web interface link on screen.